

M17 REVIEW AFTER ATTEMPT

Protected Diagnosis & Repair

THE LEARNING LOOP

- Read the route before opening assets.
- Save evidence before review.
- Changed retry only after repair.
- Keep runtime claims evidence-based.

BEGINNER-FIRST PROFESSIONAL INSTITUTE REBUILD

Preserves the authoritative 10-lesson M17 curriculum; expands teaching depth, routes, and evidence controls.

Protected review - read this first

LOCK

Open only after a genuine MetroMart attempt is saved. This book teaches diagnosis and recovery; it does not authorize replacing the original attempt.

Repair sequence

Saved attempt -> identify weak competency -> targeted teaching repair -> changed micro-proof -> validate -> reassess. If the original answer is already strong, close this book and continue.

No answer key

Do not use review to copy a polished MetroMart design. Your changed proof uses CareLink.

DE17-01 repair guide

Diagnose

Compare your saved attempt with this competency: Translate functional and non-functional requirements into an architecture decision trail. Separate what the system must do from the qualities it must protect, then record decisions in an ADR/evidence packet.

Typical failure

- Choosing products before writing requirements.
- Treating "real time" as a requirement without a number.
- Hiding budget/team limits because they feel non-technical.
- Recording a decision but no rejected alternative or revisit trigger.

Targeted teaching repair

If you feel stuck, stop drawing. Make a two-column sheet: "What must be true?" and "How will we prove it?" Convert vague words into measurable facts, then make one decision at a time.

Changed micro-proof

Only after a saved MetroMart attempt may review be opened. Repair only the weak requirement/decision link, then solve a changed CareLink micro-proof without copying the original wording.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-02 repair guide

Diagnose

Compare your saved attempt with this competency: Choose batch, streaming, or a deliberate hybrid per data flow. Explain latency, bounded/unbounded data, event time, replay, ordering, idempotency, and the condition that would make you revisit the decision.

Typical failure

- Streaming everything because it sounds modern.
- Calling a 30-minute requirement "real time."
- Ignoring duplicate delivery and external side effects.
- No replay plan, so a code defect becomes permanent data loss.

Targeted teaching repair

Write the freshness target first. Ask what breaks if data is 5 minutes, 30 minutes, or 4 hours late. Then ask whether duplicate events are safe and whether the source can be replayed.

Changed micro-proof

After saved attempt -> review -> changed CareLink decision: appointments API vs claims files. Prove why the changed freshness/correctness constraints alter or preserve your choice.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-03 repair guide

Diagnose

Compare your saved attempt with this competency: Compare warehouse, lakehouse, and combined patterns. Define raw, curated, and serving ownership so there is one understandable source of truth rather than duplicated pipelines.

Typical failure

- "Lakehouse replaces everything" as a slogan.
- Copying the same curated truth into multiple stores without ownership rules.
- Ignoring SQL concurrency/BI behavior.
- Choosing open formats but forgetting security/governance responsibilities.

Targeted teaching repair

List consumers and their access patterns. Mark which data must be replayable, which needs heavy Spark-style engineering, and which needs predictable SQL serving. Then choose the smallest coherent set of stores.

Changed micro-proof

Changed CareLink proof: decide where regulated raw claims, latest appointments state, and de-identified facility aggregates belong. Do not reuse MetroMart labels without checking constraints.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-04 repair guide

Diagnose

Compare your saved attempt with this competency: Make managed-vs-build decisions with total operational cost, control, portability, compliance, team capacity, failure responsibility, and exit strategy.

Typical failure

- Comparing only monthly cloud price.
- Assuming managed means no incidents.
- Self-hosting because it looks more "engineering."
- No exit path for a proprietary boundary.

Targeted teaching repair

Write the recurring duties: patch, backup, scale, monitor, secure, recover, upgrade, support. Put an owner beside each. If the owner column is vague, your build-yourself choice is not finished.

Changed micro-proof

Changed CareLink proof: regulation/region constraints are stronger. Re-evaluate one managed boundary; show exactly which new constraint changes the decision, or justify why it does not.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-05 repair guide

Diagnose

Compare your saved attempt with this competency: Reason about CPU, memory, I/O, network, partition skew, concurrency, quotas, small files, and downstream limits. Choose vertical/horizontal/partition scaling with load-test evidence and rollback/revisit triggers.

Typical failure

- Scaling before measuring.
- Watching average CPU while one partition is overloaded.
- Ignoring quotas/connections on the downstream sink.
- No load shape that resembles real peaks.

Targeted teaching repair

Draw the path left to right. At each hop write one capacity signal and queue/latency signal. The first place where demand exceeds sustainable service is your leading bottleneck hypothesis.

Changed micro-proof

Changed CareLink proof: acquisition doubles clinics and introduces incompatible schemas. Identify one throughput bottleneck and one schema/contract bottleneck; propose different controls.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-06 repair guide

Diagnose

Compare your saved attempt with this competency: Identify failure domains, blast radius, RTO/RPO, durable replay points, idempotency, recovery sequence, and reconciliation evidence.

Typical failure

- RTO/RPO written but no mechanism supports them.
- Backups never restored in a test.
- Replay duplicates alerts/payments.
- One shared dependency creates a hidden global blast radius.

Targeted teaching repair

Pick one failure at a time. Ask: what stops, what remains durable, how do we detect it, what is the first recovery action, what can duplicate, and what evidence proves the business state is correct?

Changed micro-proof

Changed CareLink proof: scheduling feed requires 99.95%, RTO 30, RPO 5. Design a changed recovery micro-proof and identify the evidence you would demand after failover.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-07 repair guide

Diagnose

Compare your saved attempt with this competency: Identify dominant cost drivers, unit economics, retention/tiering, autoscaling/compute windows, query/scan behavior, and guardrails with explicit trade-offs.

Typical failure

- Cutting compute until freshness fails.
- No ownership for unused storage/idle capacity.
- Ignoring scan/data-egress costs.
- Using only a total monthly bill with no unit metric.

Targeted teaching repair

Rank resources by estimated spend/usage. For the top three, write "what makes this number go up?" and one engineering control. Then test whether the control harms freshness, reliability, or compliance.

Changed micro-proof

Changed CareLink proof: target ≤ 70 unless compliance/reliability evidence justifies more. Choose one costly control you would keep and one cost you would reduce, with evidence.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-08 repair guide

Diagnose

Compare your saved attempt with this competency: Design security/governance controls around data sensitivity and trust boundaries. Use workload identity, least privilege, encryption, masking/tokenization, lineage, retention, audit, and separation of duties.

Typical failure

- One shared admin identity.
- Secrets committed to code/config.
- Masking in a dashboard while raw export stays open.
- Lineage exists but no owner/review process.

Targeted teaching repair

Start with the most sensitive field. Trace every place it can travel. At each hop ask: who needs it, why, for how long, how is access proven, and what safer representation could serve most users?

Changed micro-proof

Changed CareLink proof: patient_name, email and diagnosis_code are restricted; only de-identified facility aggregates are broad. Build one changed access/lineage proof.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-09 repair guide

Diagnose

Compare your saved attempt with this competency: For each concept: define the problem, identify when it is useful, when to reject it, operational risks, and a smaller alternative. This is architecture literacy, not five separate certifications.

Typical failure

- Calling CDC the same as streaming.
- Treating schema registry alone as a complete data contract.
- Putting Vault in the diagram without defining identity/rotation.
- Calling "teams own tables" data mesh without product/platform/governance changes.
- Using Kubernetes because containers exist.

Targeted teaching repair

For any unfamiliar concept, answer three questions only: What concrete pain does it solve? What new thing must we operate? What happens if we do not adopt it yet?

Changed micro-proof

Changed CareLink proof: incompatible schemas after acquisition make contracts more relevant; regulated secrets may change secret-management choice. Reassess without assuming every concept must become "USE."

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.

DE17-10 repair guide

Diagnose

Compare your saved attempt with this competency: Run an architecture review and defend the design under changed constraints. Communicate top risk, rejected alternatives, failure/recovery, evidence, and what decision you would revisit first.

Typical failure

- Reading every box instead of defending decisions.
- No rejected alternative.
- Claiming runtime evidence that was only simulated locally.
- Changing the whole architecture when one constraint changes.

Targeted teaching repair

Use the defense order: goal -> hard requirements -> major flow -> 3 key decisions -> biggest risk -> recovery evidence -> cost/security guardrail -> changed constraint -> revisit decision.

Changed micro-proof

If defense is weak: save attempt -> targeted review -> changed CareLink defense micro-proof -> validation -> reassessment. Do not rewrite the original attempt into a perfect answer.

Reassess

Validate the changed proof with the lesson validation criteria. If strong, record the repaired competency and continue. If still weak, return only to this competency; do not restart M17.